COS 214 – Practical 5

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1.1 Functional Requirements

Greenhouse		
FR No.	**Functional Requirement**	**Design Pattern**
1	The system must allow plants to dynamically apply different care strategies (watering, sunlight, fertilizer) based on user selection or plant type.	Strategy Pattern (Plant Care)
2	The system must allow a plant to dynamically transition between different life cycle stages – Seed, Seedling, Mature, Distressed, and Withered – based on environmental events such as germination, pruning, neglect, or recovery.	State Pattern (Plant Life Cycle)
3	The system must notify all observers (e.g. staff Observer and salesfloor Observer) whenever plant inventory is updated	Observer Pattern (Inventory Tracking)
4	The system shall allow the addition of new stock to the greenhouse inventory through a command object (Stock manager), enabling dynamic execution and potential queuing of stock addition requests without modifying existing code. This ensures seamless integration with plant creation and inventory updates as the nursery grows.	Command Pattern (Stock Management)
5	This system must create different categories of plants (such as trees, herbs, flowers) using specific factory classes that produce plant object with their corresponding care requirements	Factory Method Pattern (Plant Types)

Staff		
FR No.	**Functional Requirement**	**Design Pattern**
1	The system must allow staff to perform plant care routines where general steps are	Template Method Pattern (Plant Care Routine)

	predefined but should allow for specific care routines for specific plant types.	
2	The system should notify staff automatically whenever a plant's growth stage or health status change, so appropriate action can be taken.	Observer Pattern (Staff–Inventory Communication works with Plant Life Cycle)
3	The system must allow staff to assist customers with inquires and purchases	Chain of Responsibility Pattern (Staff Communication)
4	The system will create different types of staff members based on their assigned responsibilities	Factory Method Pattern (Staff Roles)

Customer & Sales		
FR No.	**Functional Requirement**	**Design Pattern**
1	The system shall allow customers to personalize their purchase with decorations such as pots or wrapping.	Decorator Pattern (Personalisation)
2	The system shall record transaction history to allow retrieval of past purchases.	Memento Pattern (Transaction History)
3	The system shall allow customers to choose different payment methods (e.g. EFT, E-Wallet, Cash).	Strategy Pattern (Payment Method)

Customer & Sales		
FR No.	**Functional Requirement**	**Design Pattern**
1	The system will allow customers to iterate through the list of available plants without exposing the internal structure of the collection	Iterator Design Pattern (Customer Browsing)
2	The system shall maintain an Inventory as the subject that keeps track of all plants in stock and allow observers like sales, customer service, staff to subscribe to inventory updates	Observer Design Pattern (Updating Inventory)
3	Customers interact with staff to ask for advice or help with purchases	Chain of Responsibility (Interaction with staff)

1.2 NON-FUNCTIONAL REQUIREMENTS

NFR No.	Non-functional Requirement	Quality Attribute
1	Customer and staff information should be kept safe and private	Safety
2	New plant types, payments options, or staff roles should be easy to add later	Maintainability / Extensibility
3	The system should be easy to use and understand for new users	Usability

4	All updates to plants, inventory, or sales should appear within 2 seconds	Performance / Reliability
5	The system should still work properly even if one part temporarily fails	Fault Tolerance / Reliability
6	The system can support 100 users without decreasing the uptime of the system.	Scalability